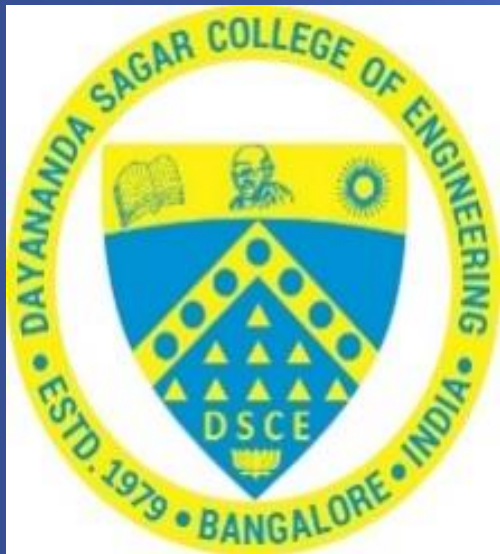


Fundamentals of Robotics and Applications

Unit 3: Robot Drive system And End Effectors



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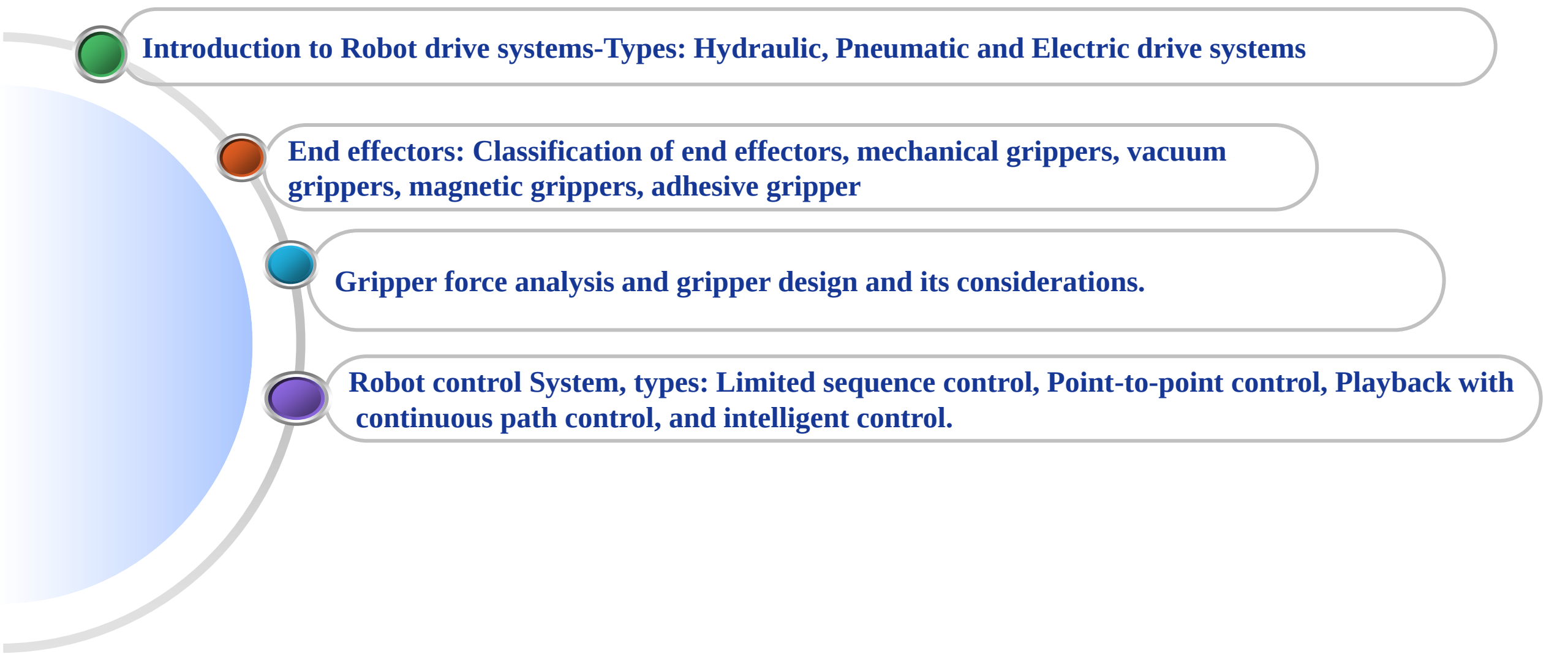
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Contents



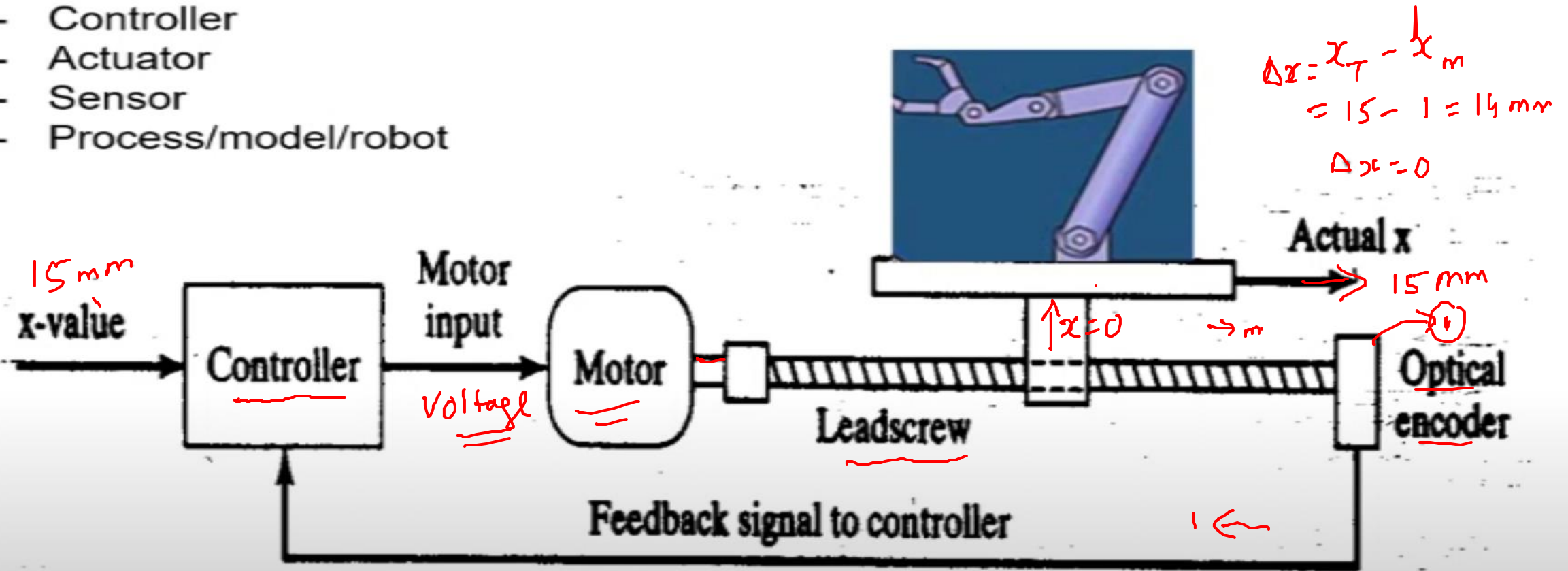
Introduction to Robot drive systems-Types: Hydraulic, Pneumatic and Electric drive systems

End effectors: Classification of end effectors, mechanical grippers, vacuum grippers, magnetic grippers, adhesive gripper

Gripper force analysis and gripper design and its considerations.

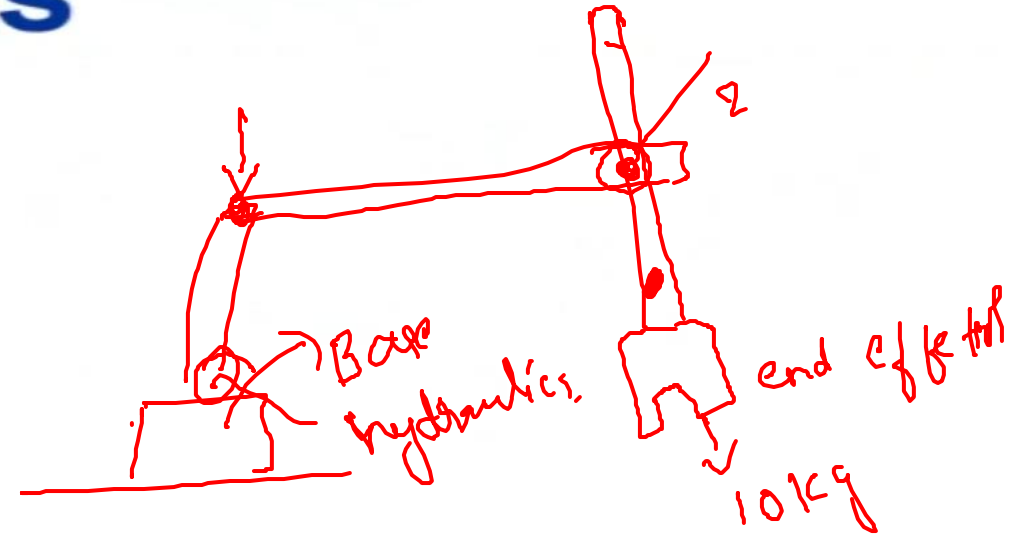
Robot control System, types: Limited sequence control, Point-to-point control, Playback with continuous path control, and intelligent control.

- Controller
- Actuator
- Sensor
- Process/model/robot



Basic elements

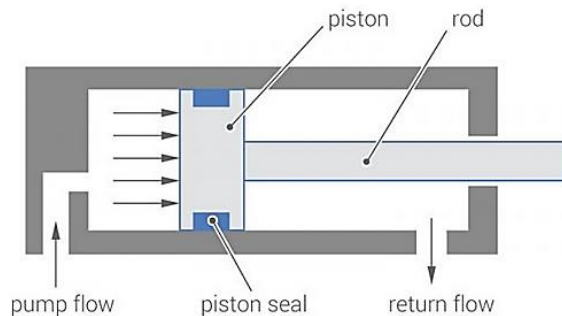
- Sensors
- Actuators
- Controllers
- System model

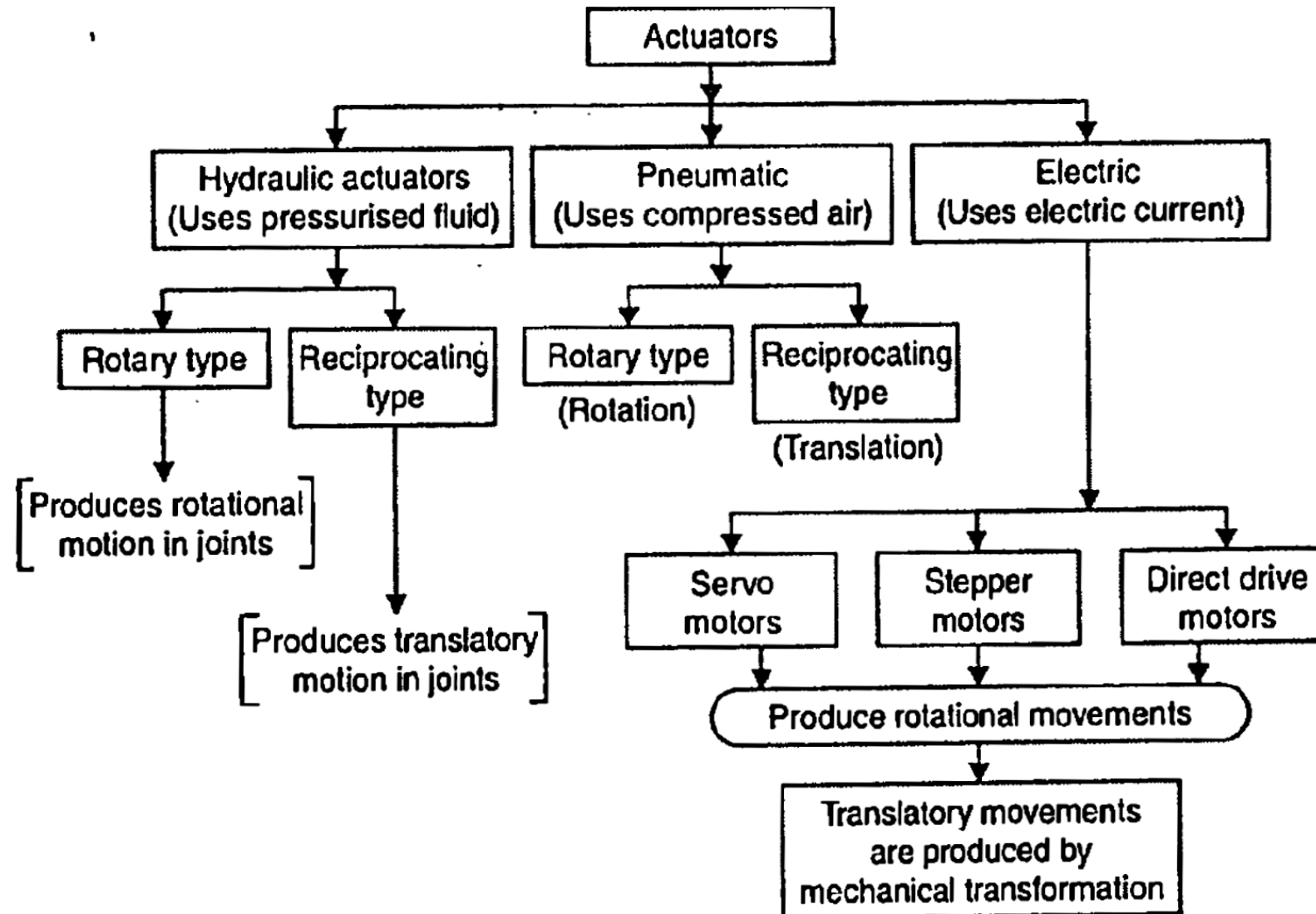


- **Basic definition of sensors and actuators :**
energy conversion from one form to another !

Robot drive systems

- The actions of the individual joints must be controlled in order for the manipulator to perform a desired move its body, arm, motion and wrist. This is provided by the drive system used to power the robot.
- The drive system is to provide a mean to control the speed and torque or power.
- Common drive systems used in robotics are **electric drive**, **hydraulic drive**, and **pneumatic drive**
- Hydraulic and pneumatic actuators generally drive prismatic joints since they produce linear motion directly. Hence, Hydraulic and pneumatic actuators are also known as linear actuators.
- **Electric motors** are suited to driving revolute joints as they produce rotation (rotary actuators).





Hydraulic Drive System:

- Hydraulic drive systems are specifically designed for larger robots or base joints for heavy load.
- It can deliver high power or speed than the electric drive systems.
- This drive system can be used for both linear and rotational joints.
- In this system, oil is used as a working fluid.
- **Principle:** an electric motor drives a pump that moves fluid from a reservoir. This causes oil to pass through the control valves and enter the actuators.
- The rotary motions are provided by the rotary vane actuators, while the linear motions are produced by hydraulic pistons.
- These systems are quite complex, costly and require maintenance.
- The leakage of hydraulic oils is considered as the major disadvantage of this drive.
- These systems deal with heavy machinery and operate smoothly, making effortless movements.

Pneumatic Drive System:

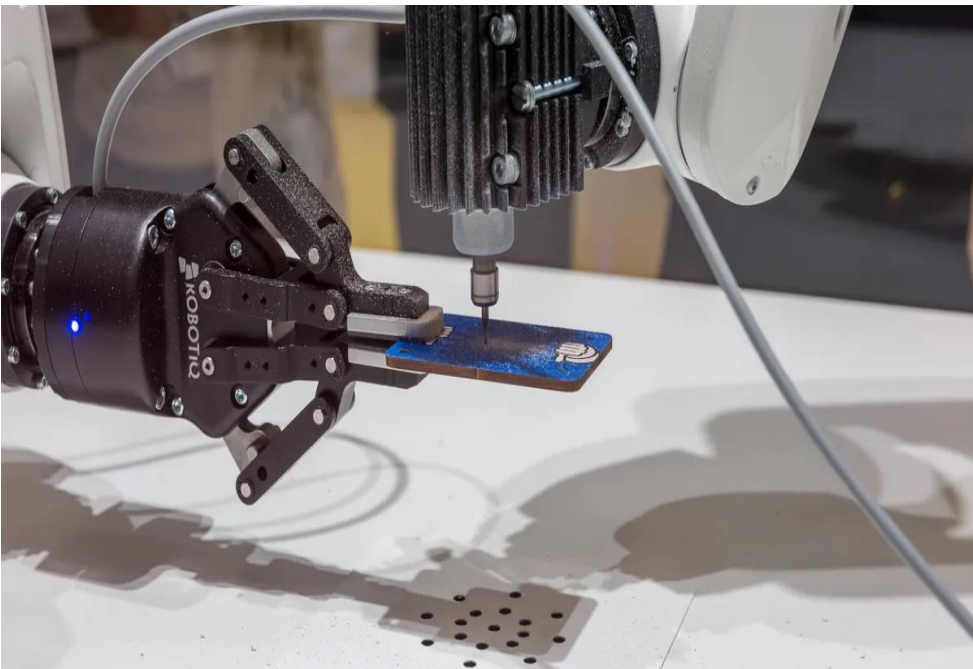
- The pneumatic drive systems are especially used for the small type robots, which have less than five degrees of freedom or grippers.
- In this system, air is used as a working fluid, hence it is also called air-powered cylinders.
- Air is compressed in the cylinder using compressor. The compressed air is used to generate the power with required amount of pressure and flow rates.
- It has the ability to offer fine accuracy and speed.
- This drive system can produce rotary movements by actuating the rotary actuators.
- The translational movements of sliding joints can also be provided by operating the piston.
- These systems are simple to construct and less expensive than hydraulic systems.
- Useful in on-off applications like pick and place robots.

Electric Drive System:

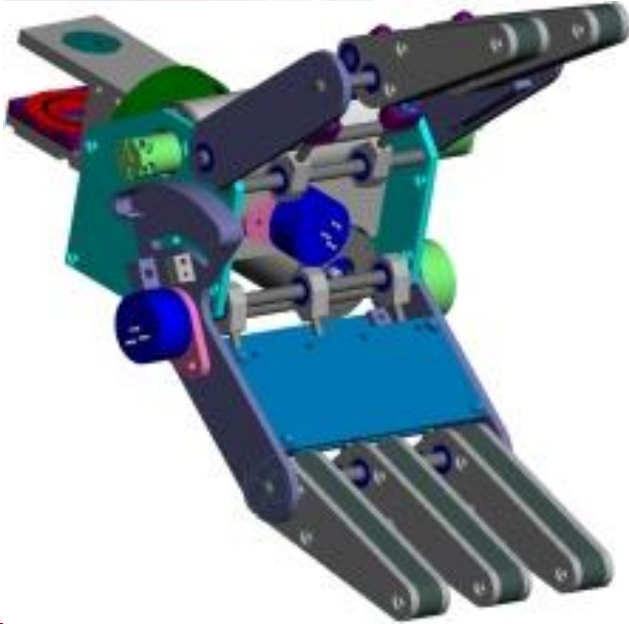
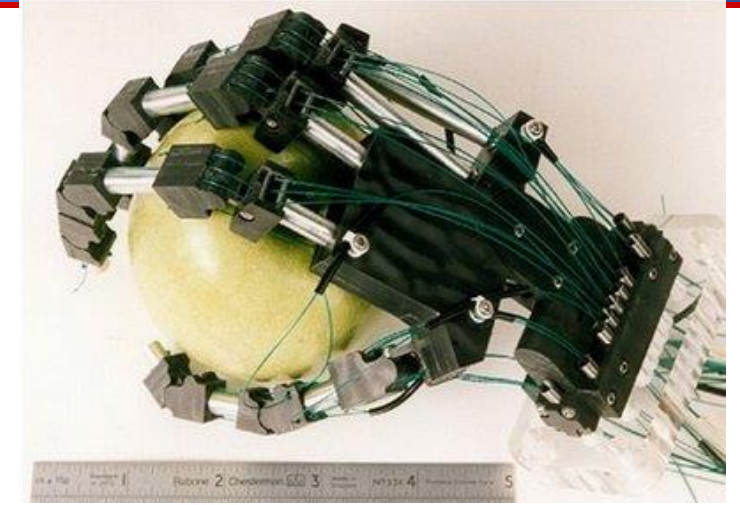
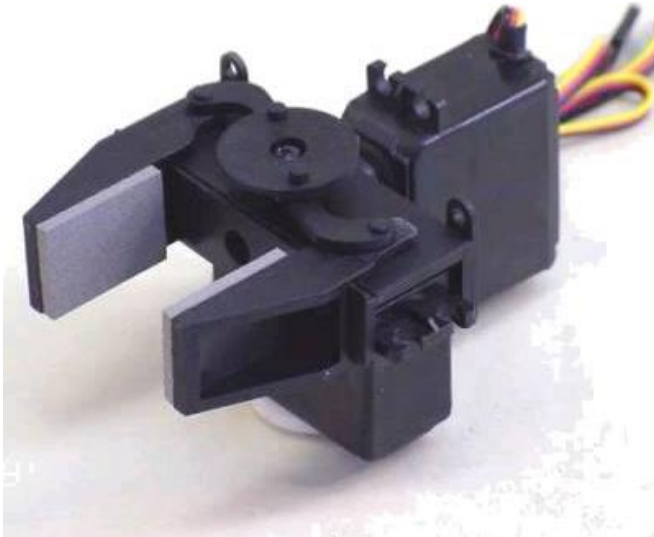
- The electric drive systems are capable of moving robots with high power or speed.
- The actuation of this type of robot can be done by either DC servo motors, BLDC or stepper motors, brushed geared motors.
- It can be well –suited for rotational joints and as well as linear joints.
- The electric drive system will be perfect for small robots and precise applications.
- They can be controlled accurately, allowing for precise movements and positioning.
- The one disadvantage of this system is that it is slightly costlier.
- Require less floor space, and their applications tend toward more precise work such as assembly.
- Ideal for tasks that require a high degree of accuracy, such as positioning a robotic arm or controlling a surgical robot.
- They can move quickly and smoothly, suitable for tasks that require fast movements.

End Effectors:

- They are tools attached to the end of a robotic arm that allows a robot to interact in an environment to perform specific tasks.
- The end effector is purchased or designed separately from the robot
- A robotic arm can use different end effectors depending on the activity: holding, cutting, welding, painting and moving different objects.



End Effectors:



Applications for End Effectors



Pick & Place



Machine Tending



Packaging & Palletizing
Processes



Assembly Tasks



Quality Testing & Inspection



Surface Finishing

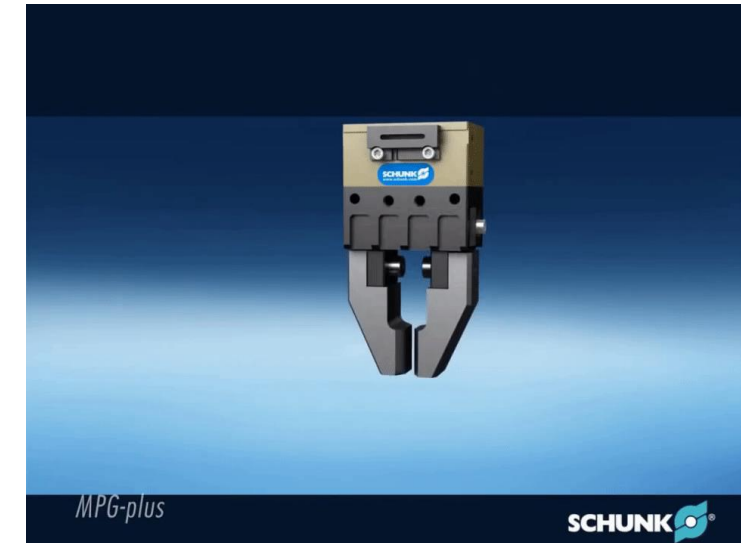
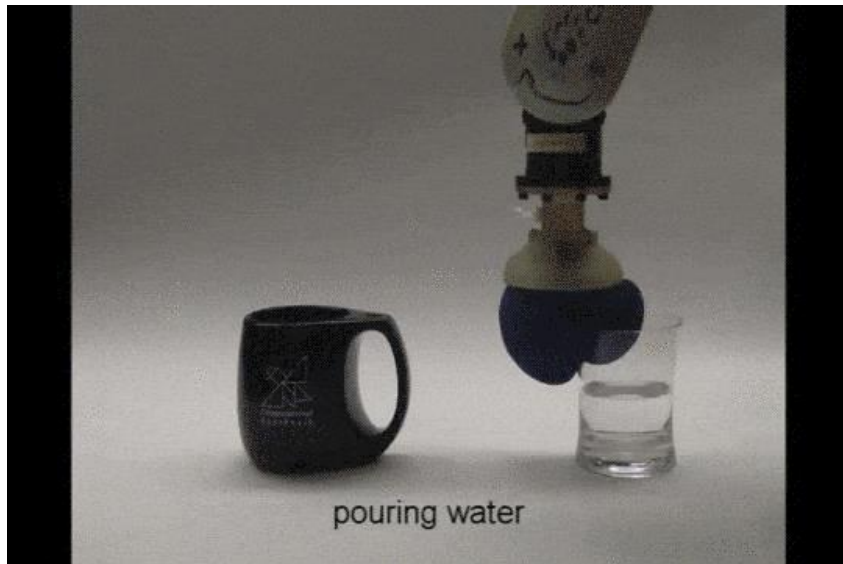
Classification of End Effectors:

End effectors are divided into two major categories:

1. **Grippers:** to grasp and manipulate objects during the working cycle.
2. **Tools:** to perform a process.

Applications

1. Grippers: pick-and-place, material handling, and assembly
2. Tools: Spot welding, arc welding, drilling, grinding, painting, etc.



Grippers:



Motors and Drives for Robotic Grippers & Tooling



DC Motor Drives



Brushless DC Motors



Integrated Motor Drives



Linear Actuators



PMDC Brush Motors



Stepper Motors



Cable / Harness Solutions



Hansen PMDC Brush
Motors

Grippers:

Grippers are end effectors used to grasp and hold objects.

Part-handling applications: machine loading and unloading, picking parts from a conveyor, and arranging parts onto a pallet.

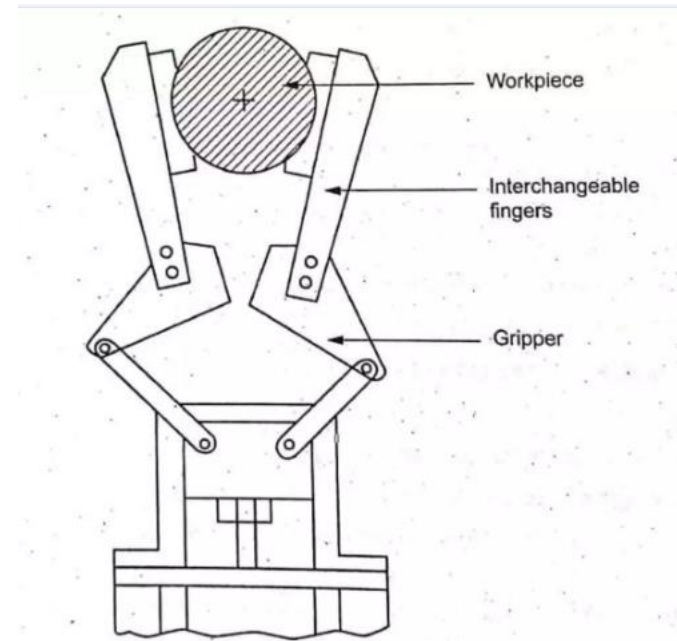
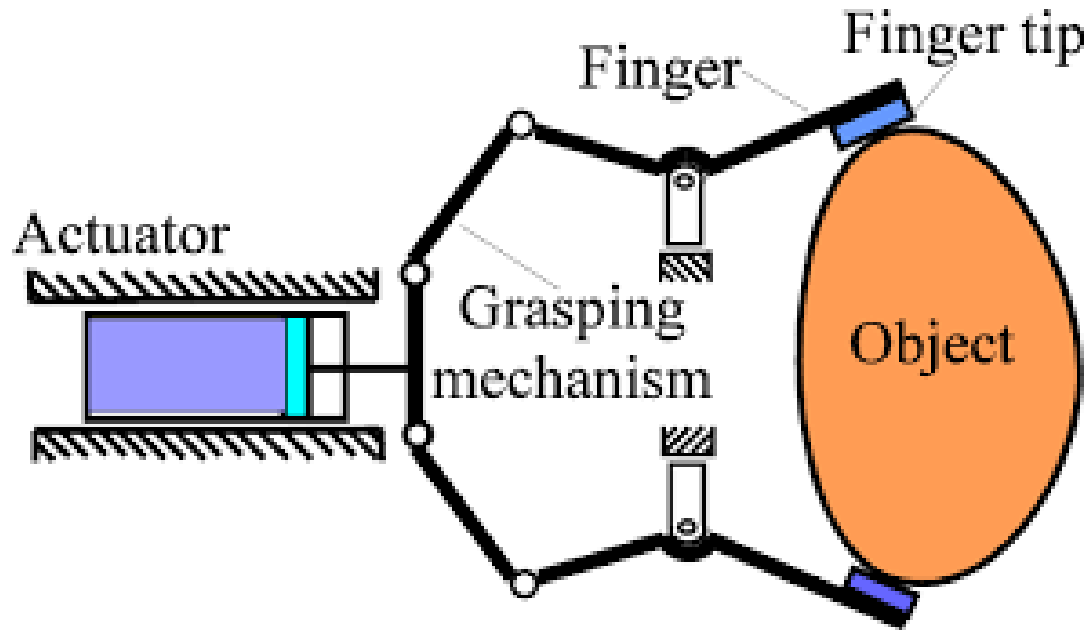
1. Mechanical grippers
2. Vacuum grippers
3. Magnetic grippers
4. Adhesive gripper

Mechanical Grippers:

- A mechanical gripper is used as an end effector in a robot for grasping the objects with its mechanically operated fingers.
- In industries, two fingers are enough for holding purposes.
- More than three fingers can also be used based on the application.
- As most of the fingers are of replaceable type, they can be easily removed and replaced.
- A robot requires either hydraulic, electric, or pneumatic drive system to create the input power.
- Mechanical grippers can be actuated by means of gear, cam links, rope & pulley, screw etc.
- The power produced is sent to the gripper to make the fingers react.
- It also allows the fingers to perform open and close actions.
- Most importantly, sufficient force must be given to hold the object.

Mechanical Grippers:

- Determine the force required to hold the object without dropping it / crushing it.
- For this reason gripper design requires much knowledge as possible of the range of items the gripper will be expected to handle.
- Mass, size, shape & strength of grippers must be taken into account.

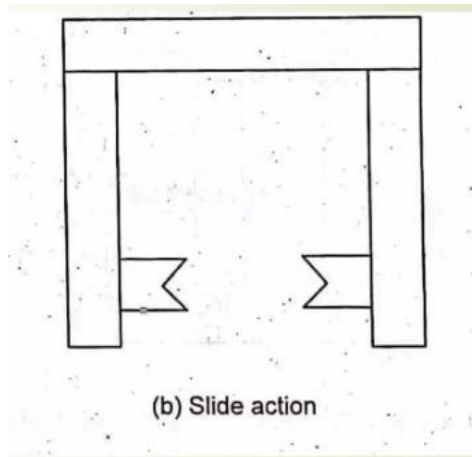


Types of Mechanical Grippers: based on primary motion of grippers

Parallel



- The jaws move in a parallel motion in relation to the gripper body.
- Comprising two opposing jaws capable of moving parallel to each other, they boast a design that lends itself effectively to grasping objects of diverse shapes and sizes.
- Accurate, simple, inexpensive, widely used.
- Be it the delicate handling of small electronic components or the secure grip of larger, irregularly shaped items, parallel jaw grippers excel in executing precise pick and place operations.
- They can be equipped with soft jaws for handling delicate items without causing harm.
- Actuation: Electrical Solenoids, Pneumatic pistons, 4bar mechanism, twin rack and pinion, wedge system, etc.



Types of Mechanical Grippers: based on primary motion of grippers

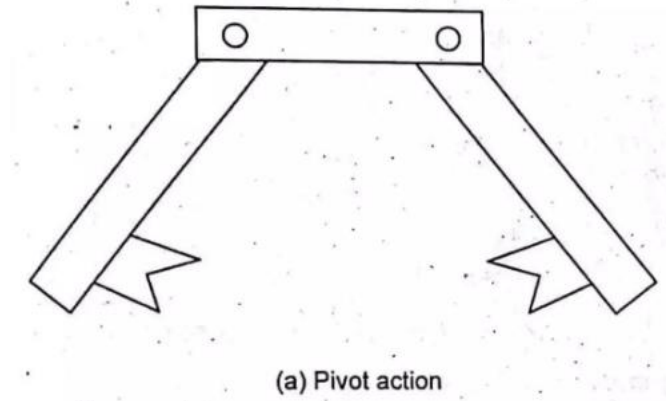
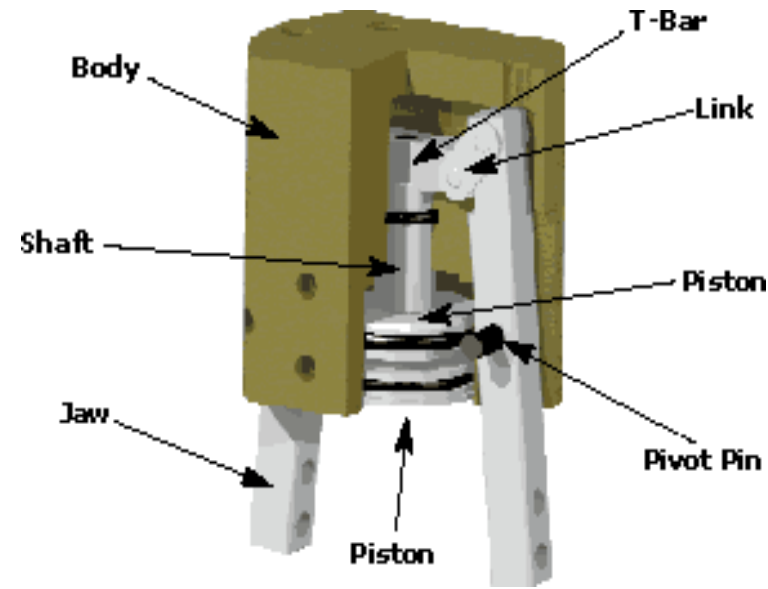


Types of Mechanical Grippers: based on primary motion of grippers



Types of Mechanical Grippers: based on primary motion of grippers

Angular



- Angular grippers: Jaws are opened/closed around a central pivot point.
- Pick up large, odd shaped objects.
- By utilizing this pivoting action, the jaws can be easily arched further out of the way to give more access to a workpiece.

Types of Mechanical Grippers: based on primary motion of grippers



Sectional diagram

The horizontally arranged pistons are pressed away from each other by compressed air.

The base jaws are opened at an angle and in a synchronized fashion by the bearing-mounted lever mechanism.

Reset is done by compression spring.

- 1 Base jaw**
for the connection of workpiece-specific gripper fingers
- 2 Lever mechanism**
for precise and synchronized gripping
- 3 Drive**
single-acting double piston system with spring return
- 4 Housing**
weight-optimized due to the use of plastics

<https://calc.schunk.com/calc-g-frame-prod/?lang=en&country=US>

Types of Mechanical Grippers: based on primary motion of grippers



Application example

Robot handling with gripper unit for loading and unloading of round plastic sleeves, are weight-optimized due to the use of plastic components.

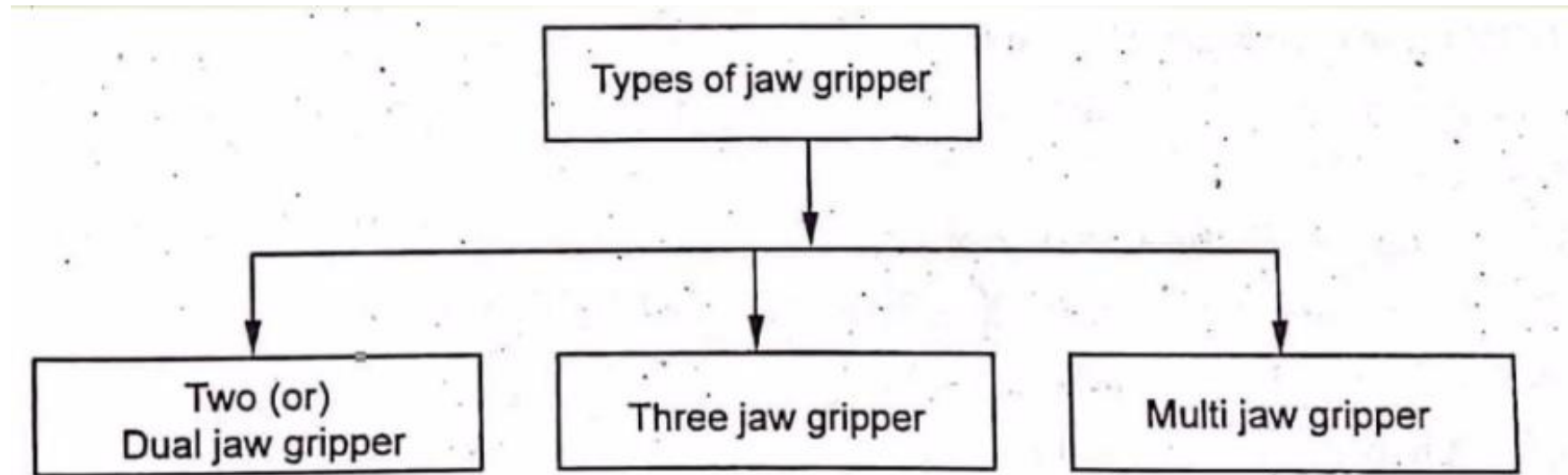
- 1 3-finger angular gripper SGW

Types of Mechanical Grippers: based on primary motion of grippers

180° Angular Air Gripper
Cam Type
MHY Series



Types of Mechanical Grippers: based on number of fingers



Types of Mechanical Grippers: based on number of fingers



Concentric/Self Centering



Individual servoed

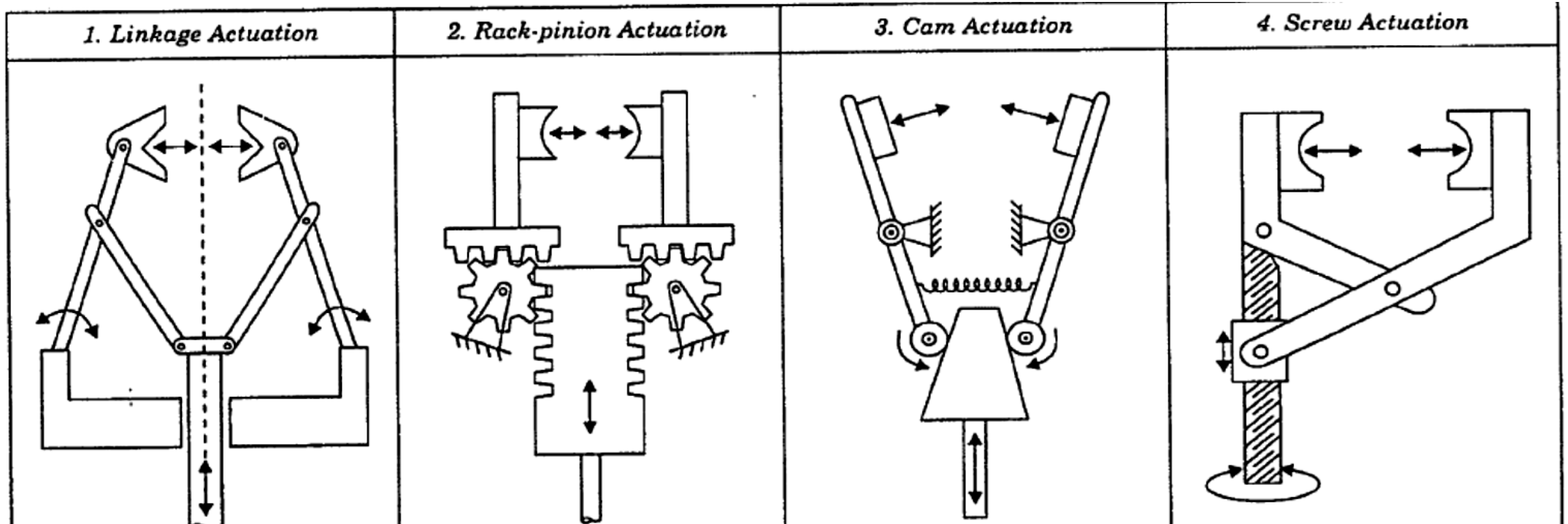


Bio mimic hand

Mechanical Grippers: Specifications

- Grip force: ranges tiny forces 0.1 lbs to over 1600 lbs (dynamic forces, moments)
- Part sizes: typically .01 to 36 inches
- Number of jaws: typically 2 to 4 jaws or fingers
- Repeatability: typically +/- .001 to .005 inches
- Cycles to failure: up to 10 million cycles
- Supporting technologies: air valves, air compressors, sensors, I/O interfaces

Mechanical Grippers: Mechanisms

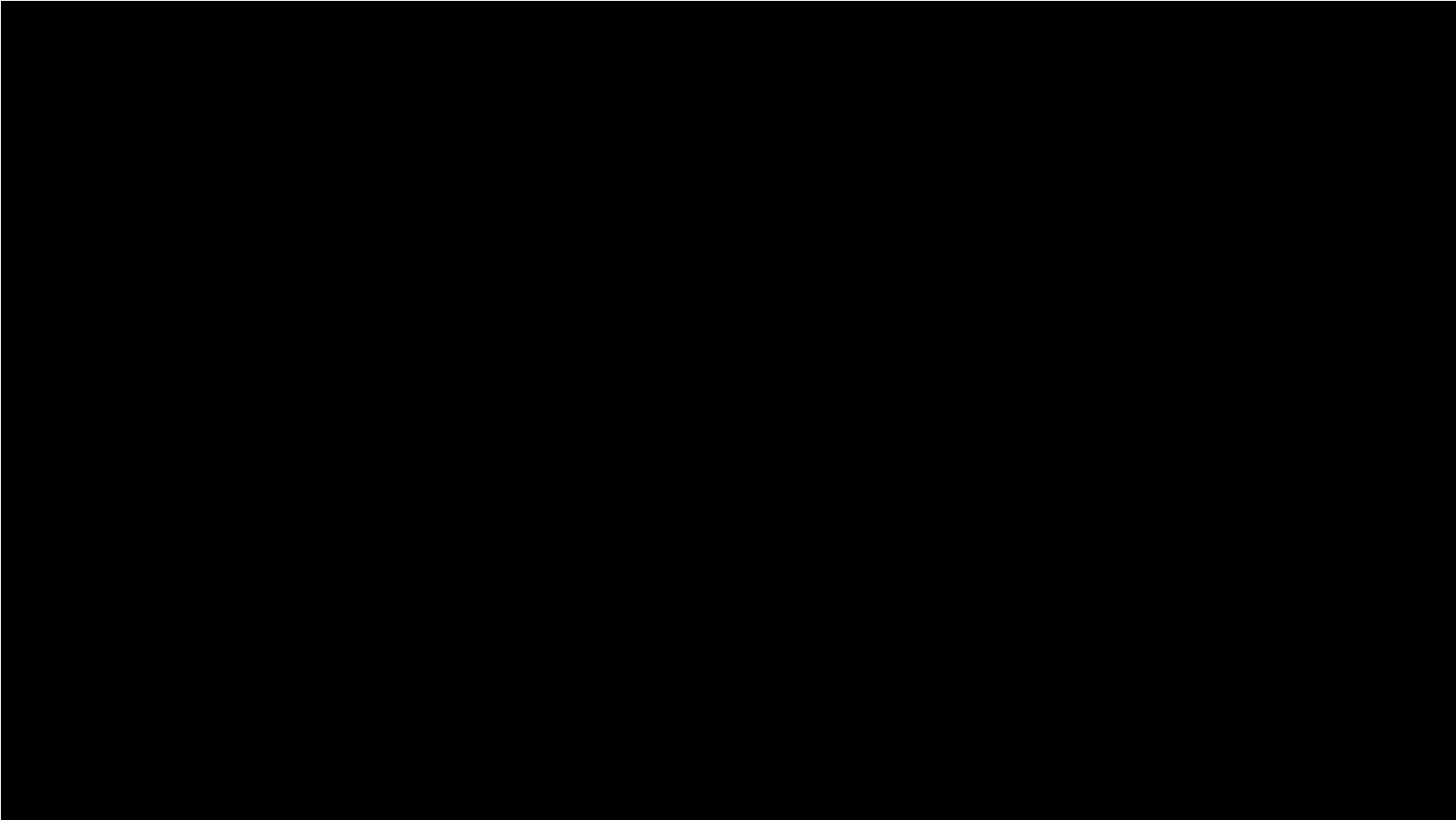


Opening width and the speed of gripping depends on the linkage configuration.

A rack actuated by a hydraulic /pneumatic cylinder operates two pivoted pinions. Pinions oscillate to move the racks.

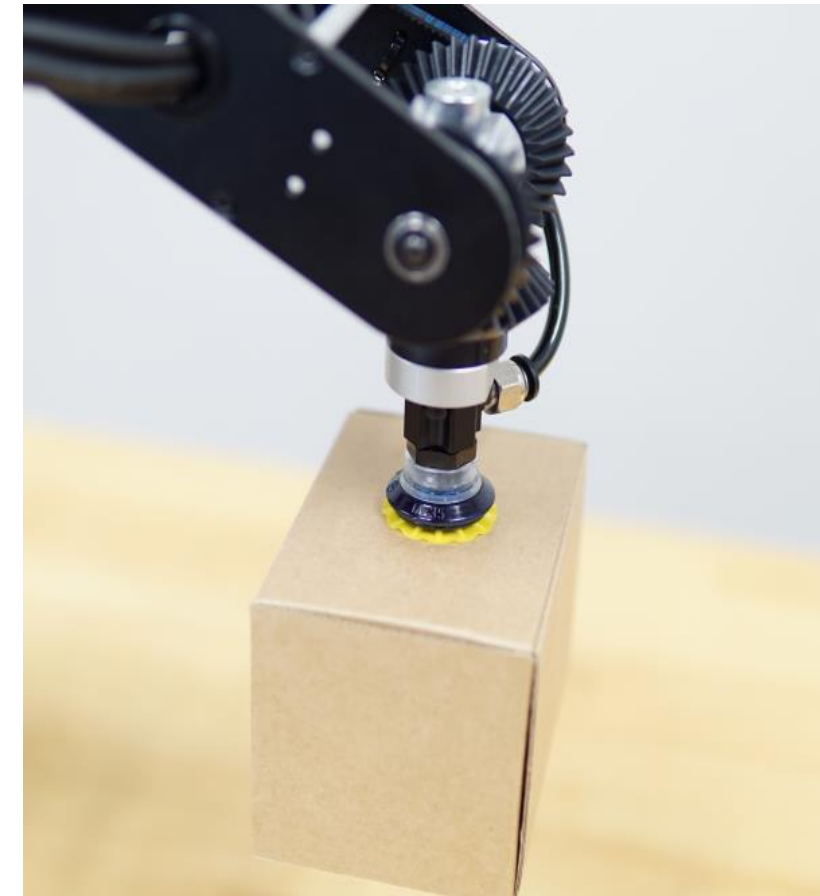
cam attached to a cylinder moves to and fro. Spring-loaded followers move on the cam profile.

A threaded block engages with a screw moving forward and backwards; the fingers attached to levers connected to the threaded block by a hinged joint open and close, providing releasing and gripping action.



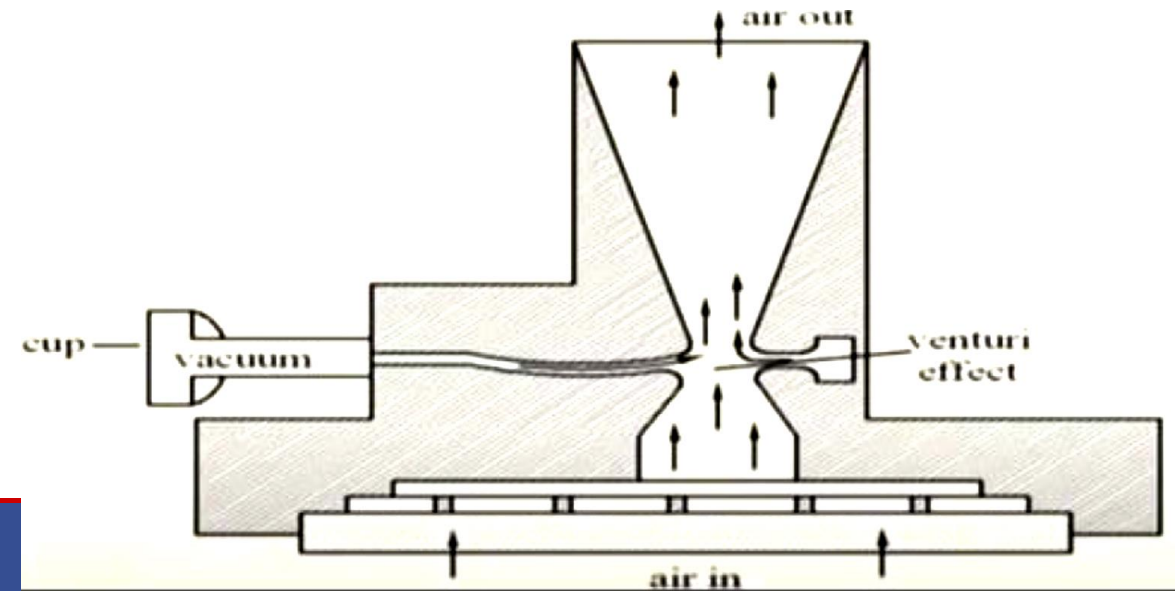
Vacuum Grippers:

- Vacuum grippers are end effectors that use suction cups to lift and manipulate objects.
- Using the difference between a vacuum and atmospheric pressure, vacuum grippers lift, hold, and move objects.
- The vacuum is created by a miniature electromechanical pump or compressed air-driven pump.
- They are widely used for robotic palletising, which is the process of stacking and arranging products on pallets for transportation or storage
- These grippers are commonly used in industries requiring precise positioning and minimal contact with the object such as the food industry (for packaging), printing, and the production of electronics.



Vacuum Grippers:

- Suction cups used in this type of robot gripper are composed of a soft material.
- Some means of removing the air between the cup and the part surface to create the vacuum is required.
- The vacuum pump and the venturi are common devices used for this purpose.
- The vacuum pump is a piston-operated or vane-driven device powered by an electric motor.
- It is capable of creating a relatively high vacuum.
- Venturi is a device that can be driven by 'shop air pressure'.
- The overall reliability of the vacuum system is dependent on the source of air pressure.





Vacuum Grippers:



Flat Suction Cups (Round)

- Handling of flat workpieces
- Low internal volume
- Fast and precise gripping



Bellows Suction Cups (Round)



- Handling of uneven and curved workpieces
- Suction bellows for height compensation
- Damping effect for sensitive workpieces



Suction Cups for Handling Sheet Metal

- Handling of metal sheets
- Handling of convex and concave workpieces
- High lateral and holding forces

Vacuum Grippers:



Suction Cups for the Packaging Industry



- Handling of cardboards, bags, plastic film and blister packs
- Use in the primary and secondary packaging sectors
- FDA compliant materials



Suction Cups for Handling Glass



- Handling of flat glass sections
- High holding and lateral forces
- Materials for leaving few marks



Suction Cups for High-Temperature Applications



- Handling of workpieces at high temperatures
- Withstands temperatures up to 600 °C

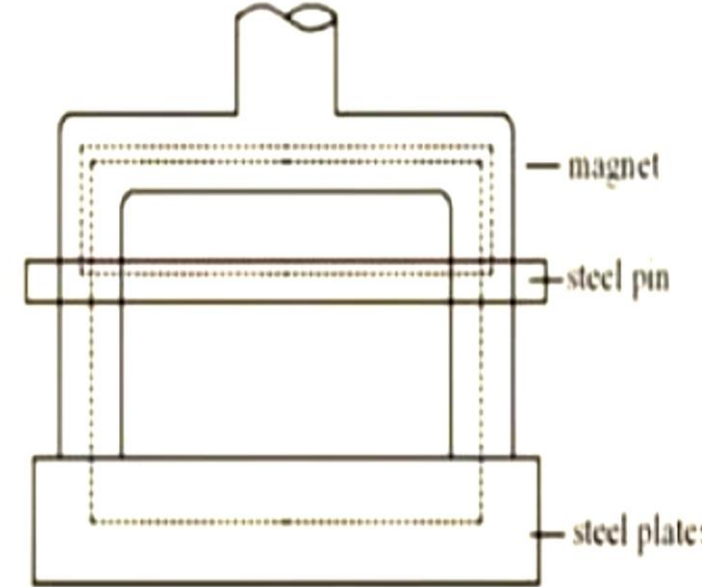
Vacuum Grippers:

Advantages

- They can handle a variety of objects
- They can lift objects from above without needing to grip them from the sides or below.
- They can distribute the payload evenly across the object's surface, avoiding deformation or damage.
- They can be customized to fit different sizes and shapes of objects by adjusting the number and position of suction cups or valves.
- They can save energy and reduce noise by using self-closing valves that only activate when needed.

Magnetic Grippers:

- Magnetic grippers can be a feasible means of handling ferrous materials (Magnetic materials).
- Can use either electromagnets or permanent magnets
 1. Pick up times are very fast.
 2. Can grip parts of various sizes
 3. Ability to handle metal parts with holes (not possible with vacuum grippers)
 4. They require only one surface for gripping.



Magnetic Grippers:

Disadvantages with magnetic grippers include;

- Residual magnetism in the workpiece, which may cause a problem in subsequent handling
- Side slippage and other errors that limit the precision of handling.
- Problem of picking thin sheet from a stack. The magnetic attraction tends to penetrate beyond the top sheet in the stack, resulting in the possibility that more than a single sheet will be lifted by the magnet



Adhesive Grippers:

- Adhesive substance performs the grasping action
- Used to handle fabrics and other lightweight materials.
- **Potential limitations** of an adhesive gripper is that the **adhesive** substance loses its **tackiness on repeated usage**, its reliability as a **gripping device is diminished** with each successive operation cycle.
- To overcome this limitation, the adhesive material is loaded in the form of a continuous ribbon into a feeding mechanism that is attached to the robot wrist.
- To release the grasp on an object, some other means, such as devices that lock the gripped object into place must be used.



Requirements of an effective gripper:

- Parts (or) items must be grasped and held without damage
- Parts must be positioned firmly (or) rigidly while being operated on
- Hands (or) grippers must accommodate parts of differing sizes (or) even of varying sizes
- Self aligning jaws are required to ensure that the load stays centred in the jaws
- Gripping (or) end effectors must not damage the part being handled
- Jaws (or) Grippers must make contact at a minimum of two points to ensure that the part does not rotate while being positioned

Gripper design considerations:

<i>Part Specification</i>	<i>Performance Specification</i>	<i>Source Specification</i>	<i>Position Specification</i>	<i>Environmental Specification</i>	<i>Material Specification</i>
• Weight of the part • Size and shape of the component • Tolerance on the part size. • Change of shape and size during processing. • Surface finish of the part. • Care for delicacy of the part to be handled.	• Co-efficient of friction between part and object. • Speed and acceleration during motion. • Accuracy and repeatability of the robot. • Interchangeability of fingers. • Memory capacity of the controller. • Control resolution of the robot. • Lead time and cycle time.	• Pneumatic: air pressure and discharge, and cylinder size. • Electrical: The power rating and specification of actuator. • Hydraulic: The oil pressure, volume flow rate and the power pack specification. • Mechanical: Linkage design and transmission design specification.	• Holding methods. • Physical difficulty • Length of fingers • The object and tool orientation. • Product design changes. • Spare parts specifications.	• Heat and temperature of object and atmosphere. • Humidity and moisture. • Dirty and safety. • Hazardness of chemicals used.	• Strength and rigidity. • Durability and fatigue factors. • Friction properties. • Factor of safety. • Design standards. • Compatibility with the environment.

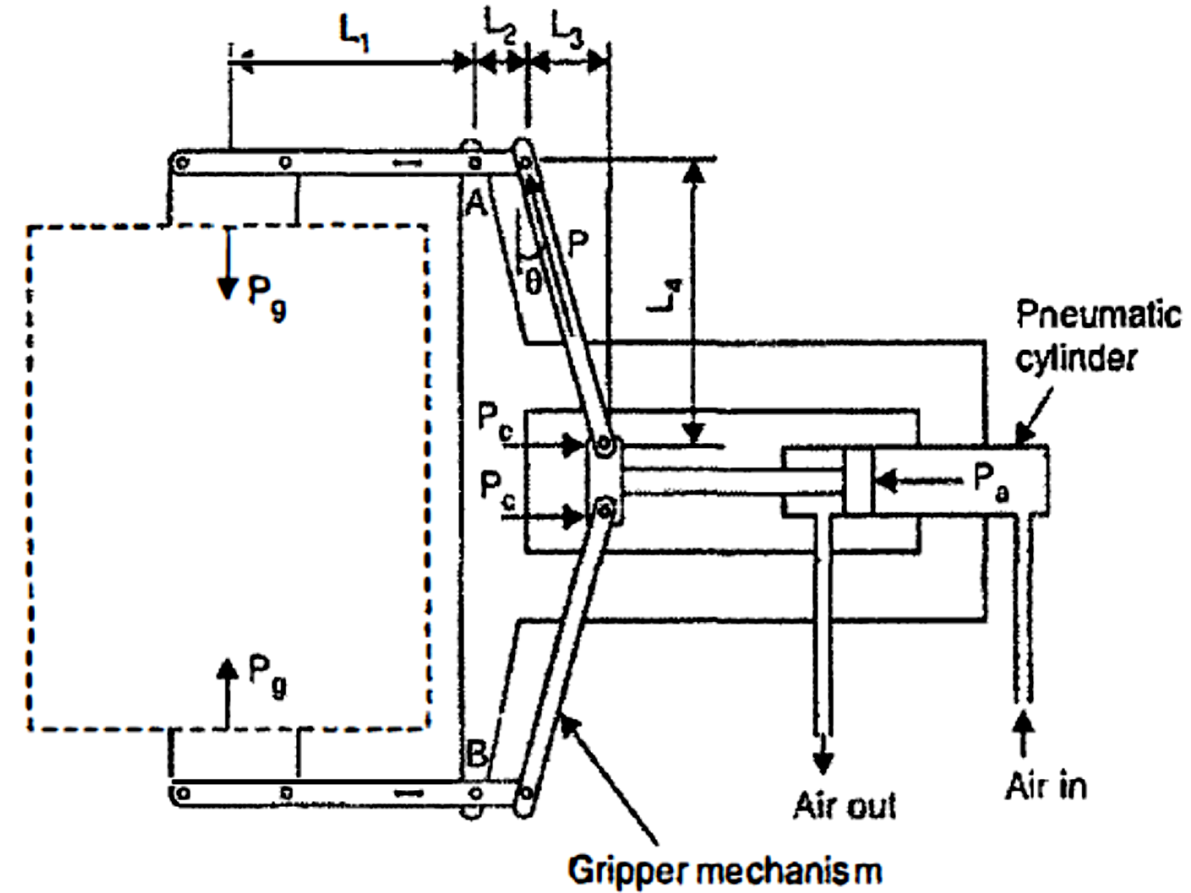
Gripper Force Analysis

- A gripper mechanism consisting of fingers, linkages frame and a pneumatic cylinder is shown in Fig.
- Air pressure supplied to the cylinder aids in actuating the fingers to grab an object with a **Gripper force** = P_g .

- m = mass of the object
- g = gravity acceleration.
- Force due to mass = $m \cdot g = W$, newtons -----(i)
- The friction between the finger pads is responsible for the gripper to hold the object exerting the force W
- The friction force is given by $f = \mu N P_g$ (ii)

μ = coefficient of friction

N = the number of fingers



Gripper Force Analysis

- Depending on the circumstances the capacity of the fingers has to be increased, incorporate a safety by a factor of safety, n .
- i.e., $F_d = \text{design force} = n.W \dots\dots\dots\text{(iii)}$

Equating equations (ii) and (iii)

$$n.W = \mu N P_g$$

$$P_g = \frac{n.W}{\mu N} = \frac{n.m.g}{\mu N}$$

If the gripper is accelerating or decelerating by 'a'

$$P_g = \frac{n.m}{\mu N} (g \pm a)$$

'a' = positive sign when accelerating down
and negative sign when accelerating up.

Gripper Force Analysis

The expression can also be written as

$$P_g = \frac{n \cdot (mg)}{\mu N} (1 \pm K_f)$$

The factor $K_f = \frac{a}{g}$ and weight of the object, $W = (mg)$

<i>Accelerating motion</i>	<i>Factor $(1 \pm K_f)$</i>
• Accelerating down	3
• Accelerating up	1
• Accelerating in horizontal direction	2

Robot control System:

Introduction

- **System:** An interconnection of elements and devices for a desired purpose.
- **Control System:** An interconnection of components forming a system configuration that will provide a desired response. Making a system behave as desired.
- **Dynamic System:** All variables that are associated with the system are functions of time.
- **System classification:**

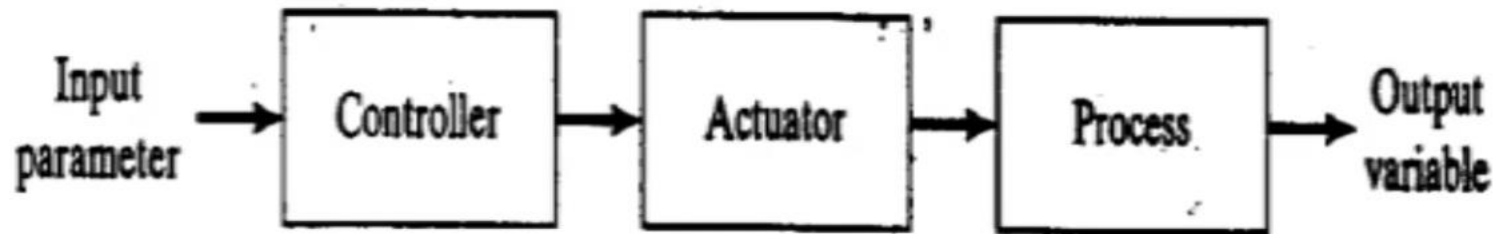
SISO: Single Input single output, MIMO: Multiple Input Multiple output

Linear, Nonlinear

Time invariant, time varying: A time invariant system is one that provides the same output for the same input irrespective of when the input is given.

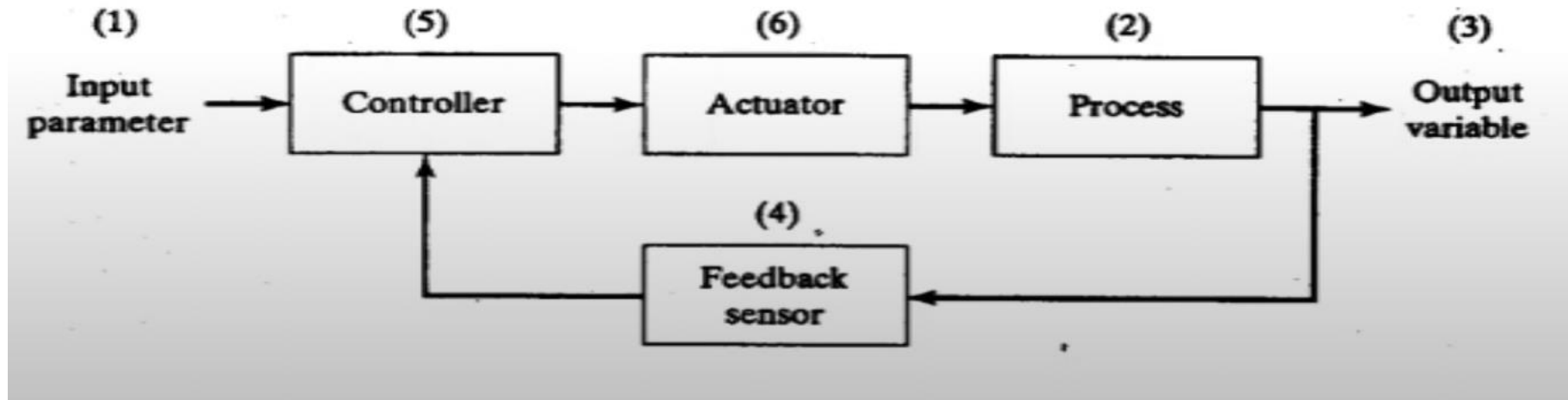
Robot control System:

- **Open Loop Control System:** Utilize a controller or control actuator to obtain the desired response. (ex: ceiling fan, TV remote, traffic lights, Conveyor belts and Automated sprinklers).
- Open-loop control systems do not use feedback from the robot's environment to adjust their actions. They rely on predefined commands and assume a predictable environment.
- These systems are typically simpler to implement and less expensive, but they are also less accurate and adaptable to changing conditions.



Robot control System:

- **Closed Loop Control System:** Utilizes feedback to compare the actual output to the desired output response. (ex: refrigerator, AC, radar tracking system, Robotic manipulators and Self-driving vehicles)
- Closed-loop control systems incorporate feedback mechanisms to monitor the robot's performance and make necessary adjustments. This allows for more accurate and adaptive behavior in dynamic environments.
- Sensors measure the robot's output and compare it to the desired output, generating an error signal. The controller then uses this error signal to adjust the robot's actions, reducing the error and achieving the desired outcome.



Robot control System:

Reactive control: (Don't think, (re)act.)

- Reactive control systems respond directly to sensory inputs, making them suitable for tasks that require quick reflexes and adaptation to real-time changes.
- These systems typically do not have internal models or planning capabilities, but they can still achieve complex behaviors by reacting to immediate stimuli.
- Examples of Reactive Control include insect-inspired robots and Reflex-based robots. Insect-inspired robots use simple sensors and reactive control algorithms to navigate complex environments and avoid obstacles. Reflex-based robots respond to sudden changes in their surroundings, such as falling objects or obstacles, by evasive actions.
- **Advantages:**
- Fast response times
- Adaptable to unpredictable environments
- Suitable for reactive tasks
- **Disadvantages:**
- Limited planning and decision-making capabilities
- Not suitable for tasks requiring long-term planning or goal-oriented behavior

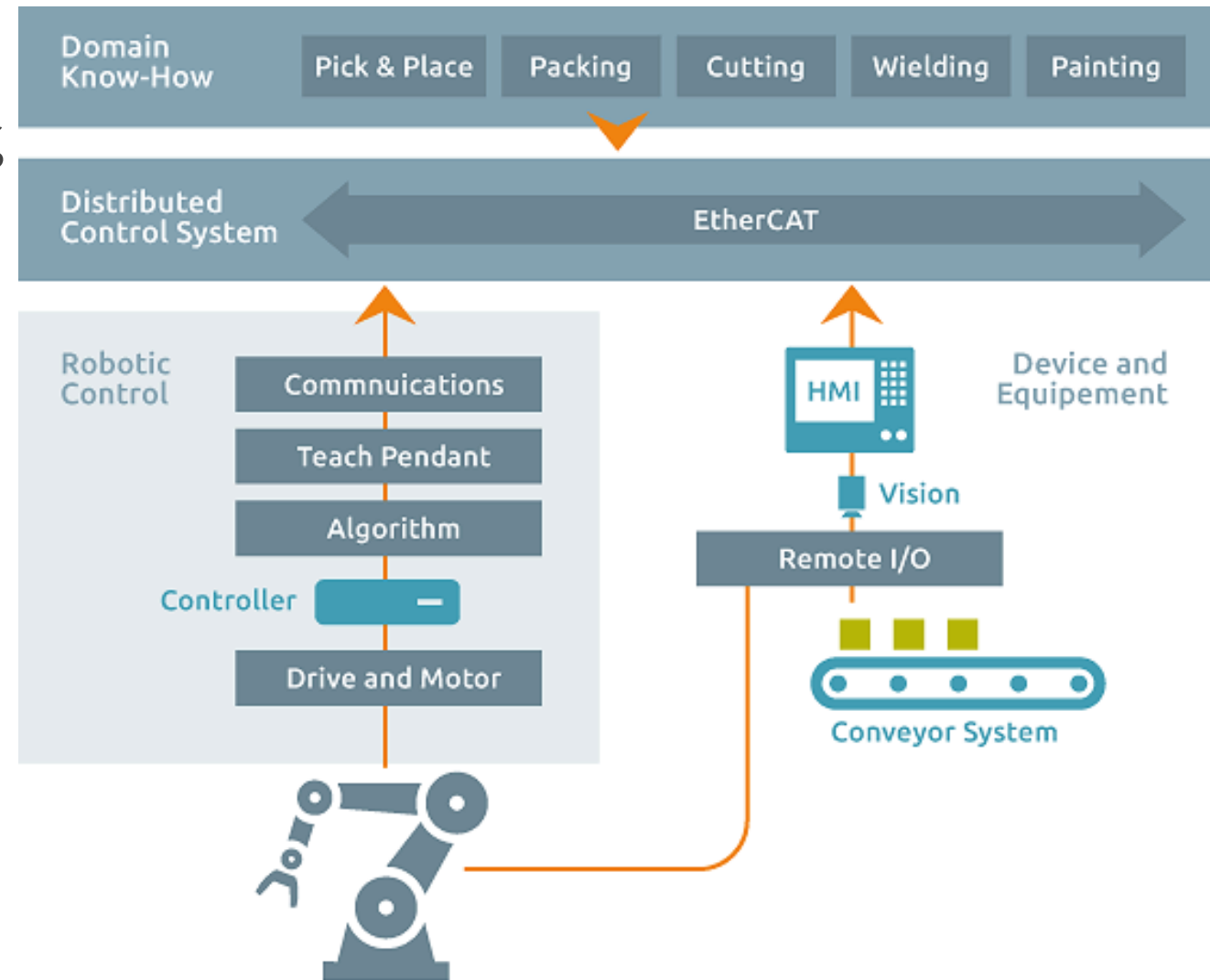
Robot control System:

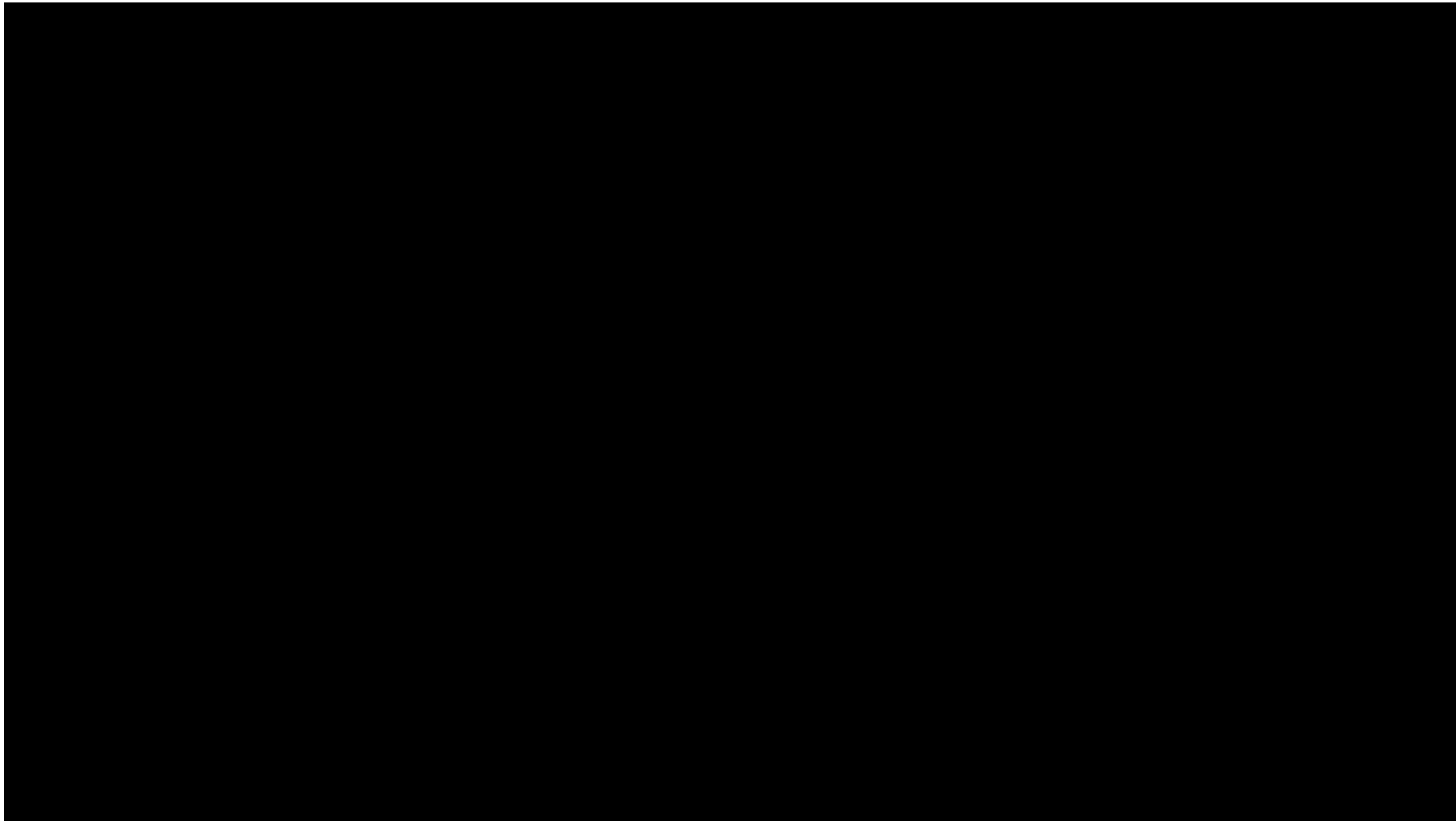
Deliberative control: (Think hard, then act.)

- Deliberative control systems use internal models and planning algorithms to generate high-level actions. They are well-suited for complex tasks that require reasoning, decision-making, and long-term planning. These systems often incorporate knowledge representation, path planning, and optimization techniques to achieve their goals.
- Examples of deliberate control include robotic surgeons and autonomous exploration robots. Robotic surgeons use preoperative planning and real-time feedback to perform delicate surgical procedures accurately. Autonomous exploration robots use internal maps and planning algorithms to navigate unknown environments, collect data, and decide their exploration path.
- **Advantages:**
 - Capable of complex planning and decision-making
 - Suitable for long-term tasks and goal-oriented behavior
 - Can handle uncertainty and incomplete information
- **Disadvantages:**
 - Computationally demanding
 - Requires accurate internal models and planning capabilities
 - Maybe less reactive to immediate stimuli

Control system components may include:

- **Controller:** PC-based system controlling the robot body.
- **Algorithms:** Application software running on the controller.
- **Teach Pendant:** Input device (HMI) enabling process control customization.
- **Communications:** Devices supporting advanced communications capabilities (e.g., EtherCAT).





Robot control System:

- Control Objectives: Control motion, Control force, Control force+motion.
- A robot manipulator consists of several mechanical links connected in series or parallel by joints.
- Actuators at each joint supply the necessary energy to move the links, while sensors measure the position of each link.
- Software controls the position and orientation of the end effector by regulating each robot joint motion to move the robot to the desired goal point.
- During the analysis of mechanical system, it is very much concern with the response of the system to certain inputs.
- The inputs may include the commands to drive the system and disturbances from the environment.
- The motions of a robot are controlled by a combination of software and hardware that is programmed by the user. Robots are classified by control method into servo and non-servo robots.

Robot control System:

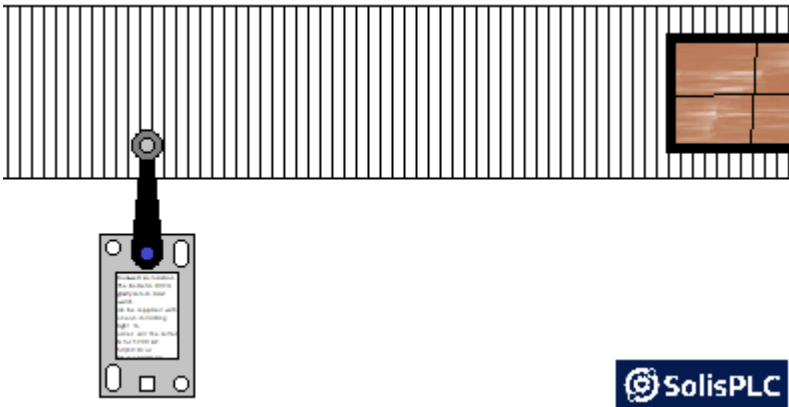
Non-Servo Controlled Robots (Limited-sequence robots):

- Non-servo control is a purely mechanical system of mechanical stops and limit switches, which are pre-programmed for specific repetitive movements.
- This can provide accurate control for simple motions at low cost.
- They do not use feedback to indicate relative positions of the joints.
- In this method of control, Individual joints can only be moved to their extreme limits of travel. (No control over the motion at the intermediate points, only end points are known)
- Pneumatic drive systems are most commonly employed.
- Non-servo robots are often referred as “endpoint,” “pick and place,” or “limited sequence” robots. These robots are used primarily for materials transfer.

Limit switches example



Limit Switch | Plunger Limit Switch - Example of a COMEPI Plunger Limit Switch



Limit Switches can be found in a number of everyday devices:

- They are used in microwaves to check that the door is closed before operation is allowed.
- They are used in elevators to notify the system that the cabin has reached the desired floor.
- Limit Switches are used in washing machine lids as a permissive to start the wash cycle.

E-Stops



Robot control System:

Non-Servo Controlled Robots:

- Programming accomplished by
 - setting desired sequence of moves
 - adjusting end stops for each axis accordingly
 - the sequence of moves is controlled by a “squencer”, which uses feedback received from the end stops to index to next step in the program
- Low cost and easy to maintain, reliable
- relatively high speed
- repeatability of up to 0.01 inch
- limited flexibility
- typically hydraulic, pneumatic drives.

Robot control System:

Servo Controlled Robots or play back robots:

- Servo control system is capable of controlling the velocity, acceleration, and path of motion, from the beginning to the end of the path.
- It uses complex control programs. These systems use programmable logic controllers (PLCs) and sensors to control the motions of robots.
- They are more flexible than non-servo systems, and they can control complicated motions smoothly.
- Sensors are used in servo-control systems to track the position of each of the axis of motion of the manipulator.
- Servo controlled robots are classified according to the method that the controller uses to guide the end-effector. These are: Point-to-point (PTP) control robot, Continuous-path (CP) control robot.

Robot control System:

Point-to-Point Control Robot (PTP):

- PTP robots are capable of performing motion cycles that consists of a series of desired point locations and related actions.
- The robot is taught each point, and these points are recorded into the robot's control unit.
- During playback, the robot is controlled to move from one point to another in the proper sequence.
- PTP robots do not control the path taken by the robot to get from one point to the next.
- If the programmer wants to exercise a limited amount of control over the path followed, this must be done by programming a series of points along the desired path.
- Applications: Component insertion, Spot-welding, Hole drilling, Machine loading and unloading, Assembly operations

Robot control System:

Point-to-Point Control Robot (PTP):

- Example: spot welding robot, the robot moves until the point to be welded is positioned between the two electrodes of the welding pistol, and welding is performed. Then, the robot moves to another point, which is again welded. This repeats until all points are welded.



Robot control System:

Continuous-path (CP) Control Robot:

- CP robots are capable of performing motion cycles in which the path followed by the robot is controlled.
- This is usually accomplished by making the robot move through a series of closely spaced points which describe the desired path.
- The individual points are defined by the control unit rather than the programmer.
- Straight line motion is a common form of CP control for industrial robots. The programmer specifies the starting point and the end point of the path, and the control unit calculates the sequence of individual points that permit the robot to follow a straight line trajectory.
- Similarly for curved path.
- To achieve CP control, controller unit storage capacity to be more.
- This usually involves the use of microprocessor.
- Applications: Typical applications include: Spray painting, Finishing, Gluing, Arc welding operations, Cleaning of metal articles, Complex assembly processes

Robot control System:

Intelligent Robots(IR):

- Growing class of industrial robot
- These IR robots not only have capability to playback a programmed motion cycle(PPT, CP) but also it can interact with environment in away of intelligence.
- The control system consists of digital computer (Programme controller).
- They can alter their programmed cycle in response to conditions that occur in the workplace.
- They can make logical decisions based on sensor data inputs from operations.
- It has the capacity to communicate during work cycle with humans and computer based systems.
- High level language- for accomplishing complex and sophisticated activities.
- Applications-assembly tasks, surgical ,under sea, nuclear, defense applications.
- IR robots made possible by availability of cheap electronics, smaller, faster processors, smart materials based actuators and sensors.

Robot control System:

Approach for control system:

1. Develop a mathematical representation for system
2. Analyse the system response
3. Design the controller